Fusion ERP — Complaint Management System (CMS)

Phase-1 One-Shot Local Code-LLM Trial Report (Qwen2.5-Coder 3B)

1. Configuration & Hardware Environment

Parameter	Recorded Value
Model & Quantization	`qwen2.5-coder:3b` (3.1B parameters, Q4_K_M, GGUF, 1.93 GB)
Installed Model Digest	`f72c60cabf6237b07f6e632b2c48d533cef25eda2efbd34bed21c5e9c01e6225`
Runtime & API Runner	Ollama `0.40.6` via `devtools/qwen_cms_trial.py` (`http://127.0.0.1:11434/api/generate`)
Hardware & OS	Apple M4 (`Mac16,12` , 10 CPU cores, 16 GB unified memory), macOS 27.0.1 (`26A434`)
Official Fusion Base SHA	`02b01fea4a67b257ea7000009f06a1561699550e` (`upstream/main`)
Trial Branch	`codex/cms-qwen-trial`
Generation Summary	20 one-shot code prompts + 1 report prompt | **408.89s** code generation wall time | **0** length truncations

2. Method & Phase-1 Scope

Every file in `modules/complaint_cms/` and `client/src/modules/complaint_cms/` was generated in a single one-shot request to the local `qwen2.5-coder:3b` model via `devtools/qwen_cms_trial.py` . Exact system/user prompts and generation options are recorded in `docs/08-delivery/cms-qwen-trial/prompts.md` , and raw payloads, raw model outputs, token counts, and SHA-256 hashes are preserved in `docs/08-delivery/cms-qwen-trial/responses/` . In accordance with Phase-1 constraints, only mechanical platform wiring and build/compiler interface adaptations were applied; no iterative business-logic re-prompting, functional workflow testing, or security hardening was performed.

3. Observed One-Shot Generation Results

Prompt	Task Label	Seconds	In Tok	Out Tok	Stop	Applied File(s)
001	Initial backend models	36.41	4098	963	`stop`	`modules/complaint_cms/models.py`
002	AppConfig and module registry	24.97	4098	575	`stop`	`modules/complaint_cms/apps.py` , `modules/complaint_cms/registry.py`
003	Pure Python domain policy and SLA constants	7.24	1185	180	`stop`	`modules/complaint_cms/domain/policy.py`
004	Public cross-module contracts	10.45	1896	237	`stop`	`modules/complaint_cms/contracts.py`
005	Complaint query service	8.22	2830	140	`stop`	`modules/complaint_cms/services/queries.py`
006	Initial complaint submission service	18.21	3025	507	`stop`	`modules/complaint_cms/services/complaints.py`
007	Complaint workflow actions service	53.92	2642	1894	`stop`	`modules/complaint_cms/services/actions.py`
008	Scheduled SLA and auto-close events service	13.98	1177	470	`stop`	`modules/complaint_cms/services/events.py`
009	Celery tasks and beat schedule	13.33	1333	297	`stop`	`modules/complaint_cms/tasks.py` , `modules/complaint_cms/schedule.py`
010	DRF API serializers	23.96	1196	747	`stop`	`modules/complaint_cms/api/serializers.py`
011	Employee and action API views	32.67	4450	424	`stop`	`modules/complaint_cms/api/views.py`
012	Configuration and record API views	15.25	2354	250	`stop`	`modules/complaint_cms/api/configuration.py`

Prompt	Task Label	Seconds	In Tok	Out Tok	Stop	Applied File(s)
013	API URL registration	8.88	875	148	`stop`	`modules/complaint_cms/api/urls.py`
014	Frontend TypeScript interfaces	13.25	1901	370	`stop`	`client/src/modules/complaint_cms/types.ts`
015	Frontend typed API client	13.28	1374	291	`stop`	`client/src/modules/complaint_cms/api.ts`
016	Frontend route registration	6.61	670	154	`stop`	`client/src/modules/complaint_cms/routes.tsx`
017	Initial complainant page	38.36	1218	1164	`stop`	`client/src/modules/complaint_cms/pages/MyComplaintsPage.tsx`
018	Initial authority work queue page	24.15	1179	743	`stop`	`client/src/modules/complaint_cms/pages/ComplaintQueuePage.tsx`
019	Initial complaint configuration and audit page	30.27	1172	899	`stop`	`client/src/modules/complaint_cms/pages/ComplaintAdminPage.tsx`
020	Generate module documentation	15.48	843	382	`stop`	`modules/complaint_cms/README.md`

4. Build & Integration Check Results

Recorded Check	Command	Exit Status	Seconds
Django model registration check	`.venv/bin/python manage.py check`	1 (INITIAL)	0.55
Django model registration after import adaptation	`.venv/bin/python manage.py check`	1 (INITIAL)	0.60
Django model registration after import adaptation	`.venv/bin/python manage.py check`	1 (INITIAL)	0.31
Django model registration after import adaptation	`.venv/bin/python manage.py check`	1 (INITIAL)	0.62
Django model registration after import adaptation	`.venv/bin/python manage.py check`	0 (PASS)	0.32
Generate schema migration from registered Qwen models	`.venv/bin/python manage.py makemigrations complaint_cms --noinput`	0 (PASS)	0.37
backend-python-compilation	`.venv/bin/python -m compileall -q modules/complaint_cms`	0 (PASS)	0.03
isolated-cms-schema-install	`.venv/bin/python manage.py migrate --noinput`	0 (PASS)	0.36
schema-model-drift	`.venv/bin/python manage.py makemigrations --check --dry-run`	0 (PASS)	0.32
isolated-module-registration	`.venv/bin/python manage.py seed_modules`	1 (INITIAL)	0.29
isolated-module-registration	`.venv/bin/python manage.py seed_modules`	0 (PASS)	0.50
cross-module-schema-boundaries	`.venv/bin/python ops/checks/no_cross_module_fk.py`	0 (PASS)	0.15
registered-import-boundaries	`.venv/bin/lint-imports`	0 (PASS)	0.15
navigation-route-registration	`.venv/bin/python ops/checks/nav_matches_routes.py`	0 (PASS)	0.15
frontend-first-typecheck	`npx tsc --noEmit`	2 (INITIAL)	2.06
frontend-final-typecheck	`npx tsc --noEmit`	2 (INITIAL)	1.59
frontend-final-typecheck	`npx tsc --noEmit`	0 (PASS)	1.49
frontend-production-build	`npm run build`	0 (PASS)	3.33

5. Mechanical Integration Corrections

- Registered `modules.complaint_cms` in `config/settings/base.py` (`DOMAIN_MODULES`), mounted `api/v1/complaints/` in `config/urls.py`, merged `BEAT_SCHEDULE` and `TASK_ROUTES` in `config/celery.py`, added `complaint_cms` to `.importlinter` domain/module/layer contracts, and registered `complaint_cms` in `client/src/app/registry.ts`. Exact diff: `integration-registration.patch`.
- Created empty Python package `__init__.py` files for `modules/complaint_cms/`, `domain/`, `services/`, `api/`, and `migrations/` as mechanical package scaffolding. Application logic remains model-generated.
- Prompt 001 (`modules/complaint_cms/models.py`): added missing `from dataclasses import dataclass` import for the placeholder `Principal` dataclass and increased `status` / `update_type` / `old_status` / `new_status` `max_length` from `10` to `20` to satisfy Django `fields.E009` (`'In Progress'` is 11 chars and `'StatusChange'` is 12 chars). Model fields, choices, and defaults were preserved unchanged.
- Prompt 002 (`modules/complaint_cms/registry.py`): aligned `NAV_ITEMS` dictionary keys (`code`, `label`, `icon`, `to`, `required_permission`, `sort_order`) with the required `seed_modules` / `nav_matches_routes.py` contract and granted `complaint_cms.task.view` to authority roles so every declared permission has a holder.
- Prompt 004 (`modules/complaint_cms/contracts.py`): added missing `from dataclasses import dataclass` import and aliased `UserScopedModel as UserRef` to match the generated query without changing contract logic.
- Prompts 005–008 (`modules/complaint_cms/services/*.py`): adapted single-dot relative imports (`.models`, `.domain.policy`, `.registry`) to parent package relative imports (`..models`, `..domain.policy`, `..registry`), mapped exception imports to `core.api.exceptions`, and replaced `pytz` with `django.utils.timezone` in `events.py`. All query filters, submission logic, workflow action branches, and scheduled event bodies were preserved unchanged.
- Prompt 009 (`modules/complaint_cms/tasks.py`, `schedule.py`): added missing `@shared_task` decorator to `cms_process_sla` and removed unused `pytz` import in `schedule.py`.
- Prompts 010–013 (`modules/complaint_cms/api/*.py`): adapted single-dot model/service imports to `..models` / `..services`, imported `HasModuleGrant` and `HasPermission` from `fusion_auth.permissions`, replaced invalid `CharField(choices=...)` with `ChoiceField(choices=...)` and `null=True`/`blank=True` with `allow_null=True`/`allow_blank=True` on DRF serializer field overrides, wrapped `record_list_view` in `RecordListView(APIView)` to match the generated `urls.py` import, and stripped trailing slashes in `urls.py` to match the platform API convention.
- Prompts 014–019 (`client/src/modules/complaint_cms/**`): added `export` keywords to `api.ts` functions and re-exported `./types`, added `Navigate` import in `routes.tsx`, and adapted React Query v5 object-form calls (`useQuery({ queryKey, queryFn })`, `useMutation({ mutationFn })`, `invalidateQueries({ queryKey })`), Mantine v7 component props (`gap`, `justify`, `highlightOnHover`, `value` on `Pagination`), and named imports (`{ PageHeader }`, `{ errorMessage }`) so `tsc --noEmit` and `npm run build` succeed without altering generated UI structure or API payloads.

6. Final Application File Inventory & SHA-256 Hashes

File Path	Lines	Bytes	SHA-256 Digest
`client/src/modules/complaint_cms/api.ts`	44	1350	`8384839477c4a52b7dd983ea9d5bca106936d4b9d0c956ad628d8d3752c55cda`
`client/src/modules/complaint_cms/pages/ComplaintAdminPage.tsx`	131	4414	`579e7684434ae18724184254fb0ec44de7a713f8bd08e1ce81e2d55683217ae8`
`client/src/modules/complaint_cms/pages/ComplaintQueuePage.tsx`	92	3382	`e4bedc84e89c0a2e5fca8b099843c93fccf0bd39a5f62a70067ec9a0ba32e0cb`
`client/src/modules/complaint_cms/pages/MyComplaintsPage.tsx`	152	5098	`e9c5ea70ee14c33d116973ac8a929b86124c7f42ee948a82948d856ac8448679`
`client/src/modules/complaint_cms/routes.tsx`	13	582	`3c6e0e12f49db469a1bbec658fc106362ceaa0dc9bbedbb1df79c7c2b88065d8`
`client/src/modules/complaint_cms/types.ts`	78	1606	`d4f55f943e466683a84d796922c7b6f0a8fd259050d49cf07777fbfede11c592`
`modules/complaint_cms/README.md`	39	1898	`90f888ada78c254de1372f27c1f515f1e8464bea286f4a4a3849d133440cbe6f`
`modules/complaint_cms/__init__.py`	0	0	`e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855`
`modules/complaint_cms/api/__init__.py`	0	0	`e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855`

File Path	Lines	Bytes	SHA-256 Digest
`modules/complaint_cms/api/configuration.py`	33	1507	`e495efa82195edf809f972957a099273b6f052f29ead868a1e563d9fc4482659`
`modules/complaint_cms/api/serializers.py`	101	3806	`c1dc80b08ef439b74fa18755547af2a1f5f637a64500c014614cb1578dd99569`
`modules/complaint_cms/api/urls.py`	20	703	`e0adeaa5368fd01b1180d37e2b5215630f0fdb9ac2cd2009ff80c2312ad39105`
`modules/complaint_cms/api/views.py`	51	2204	`92bbe9c1b5c6e78ea842c9fbe09f4c6f6e99da92854e17b97d74561c2d981021`
`modules/complaint_cms/apps.py`	7	240	`0d1f48e8da9a44814cb8d4596c4024fa057d61db78761370dbe2febee6d22d7a`
`modules/complaint_cms/contracts.py`	34	1126	`f1db0511c28a512ffe1fc143b4e995ef4a0f0f12ab4b9ab5fbd3e7afb3e13a19`
`modules/complaint_cms/domain/__init__.py`	0	0	`e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855`
`modules/complaint_cms/domain/policy.py`	29	642	`07f30f472615b6c9e0542c158a6db0b1582255dd38479925b975e7036a8e8bc5`
`modules/complaint_cms/migrations/0001_initial.py`	155	8275	`d978da7ca0c458e012332be57c68ab5140f86a72b5f8835b3dc54fd3200e65be`
`modules/complaint_cms/migrations/__init__.py`	0	0	`e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855`
`modules/complaint_cms/models.py`	123	4639	`0644e06df3d2fd05f6a4804307fcfbbf001d75e633825110c2eb5ce0fd98d933`
`modules/complaint_cms/registry.py`	35	1998	`aedb7a730f1abebc40ff939f7ac328702309ef03818f2b7e21027bfcc57514c2`
`modules/complaint_cms/schedule.py`	19	572	`a02d0b019a752f42316a61d95461f2a14c3f9278fce511e275a96b8889dccd8d`
`modules/complaint_cms/services/__init__.py`	0	0	`e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855`
`modules/complaint_cms/services/actions.py`	260	11865	`814af36f0fe030ad2f8a57ce592332982124752ed5b51108259d22a0b02f7dec`
`modules/complaint_cms/services/complaints.py`	86	2742	`f5de32caa7cf9363f2dc24b4c625e4f2cbe74a9b5b2d70316e41ad941b7a6bee`
`modules/complaint_cms/services/events.py`	57	2351	`7c34888ee24418aae0c3be16f7310182b06f24d4d0e5bc9d3d9a311af0d31c12`
`modules/complaint_cms/services/queries.py`	19	752	`ab3c8ff1f712331ea7a6f7f70fd8ae0053e4a03a926e549a41623993f31a5b35`
`modules/complaint_cms/tasks.py`	11	296	`8e2d762c0e32a274d7cc0d9685568e82abf8399ade66dd1b62d54eb6a34c1b68`

7. Reproduction Commands

With Ollama running locally and `qwen2.5-coder:3b` installed:

```
python3 devtools/qwen_cms_trial.py generate docs/08-delivery/cms-qwen-trial/jobs/05-queries.json
```

```
.venv/bin/python devtools/cms_trial_check.py --label django-check -- .venv/bin/python manage.py check
```

```
.venv/bin/python devtools/cms_trial_check.py --label cms-schema --isolated -- .venv/bin/python manage.py migrate --noinput
```

```
.venv/bin/python devtools/cms_trial_check.py --label frontend-build --client -- npm run build
```